

Cross-Lingual Persona Consistency in Instruction-Tuned Large Language Models

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ABSTRACT

We investigate whether persona consistency in instruction-tuned large language models (LLMs) degrades when the interaction language shifts from English to German or Italian, and whether certain personas or topic categories are more vulnerable to this degradation. Two models of different scales — LLAMA-3.1-8B and LLAMA-3.3-70B — are evaluated under four persona prompts (*Neutral*, *Pessimist*, *Scientist*, *Teenager*) across 20 open-ended questions in English, German, and Italian, spanning four thematic categories. We combine automatic metrics (sentiment polarity, lexical formality, multilingual embedding consistency) with an LLM-as-judge rubric assessing persona adherence, fluency, and drift resistance. Our central finding is that persona drift is primarily a *register* problem rather than a *content* problem: models preserve semantic meaning across languages but lose stylistic precision, particularly in Italian and for informal or technically demanding personas. *Neutral* and *Pessimist* are robustly maintained (judge adherence ≥ 4.75), while *Scientist* and *Teenager* degrade substantially (≈ 3.5) despite moderate embedding consistency. Italian shows the highest drift (judge drift resistance 3.68 vs. 4.81 in English), while DE $\leftrightarrow$ IT is the most stable language pair (0.832) — suggesting English responses are the divergent anchor rather than Italian. Abstract topic categories further compound persona instability. These findings have direct implications for the design of register-aware multilingual dialogue systems.

1. INTRODUCTION

Large language models are increasingly deployed in multilingual settings where users interact in languages other than English [Ahuja et al., 2023]. While significant progress has been made in cross-lingual transfer for core NLP tasks, the behavioural properties induced by *system-level instructions* — such as persona, tone, and communication style — remain understudied across languages.

Persona conditioning via system prompts is a standard technique in dialogue system design: a customer service agent, pedagogical tutor, or creative writing assistant must maintain a coherent character across turns. What is less understood is whether models preserve persona-specific traits — sentiment register, lexical formality, stance, and self-presentation — when the conversation language switches from English to a typologically different or lower-resource language. We define *persona consistency* as the degree to which a model maintains the same sentiment polarity, lexical register, and semantic framing across languages when conditioned on the same persona prompt.

This paper investigates three questions:

1. Does cross-lingual persona consistency differ across persona types, and is it primarily a register or content problem?

2. Is Italian more susceptible to persona drift than English or German, and does this vary by topic category?
3. Does model scale affect consistency, and does this effect interact with persona type?

We evaluate two Llama models (LLAMA-3.1-8B and LLAMA-3.3-70B) under four persona prompts across 20 open-ended questions in English, German, and Italian, using automatic metrics (sentiment polarity, lexical formality, multilingual embedding consistency) and an LLM-as-judge rubric. Our central finding is that persona drift is primarily a *register* problem: models preserve semantic content across languages but lose stylistic precision, especially in Italian and for informal or technically demanding personas. *Neutral* and *Pessimist* are robustly maintained, while *Scientist* and *Teenager* degrade substantially. Contrary to expectations, English responses are the divergent anchor: DE $\leftrightarrow$ IT is the most stable language pair.

2. RELATED WORK

Multilingual LLMs. Multilingual models such as mBERT [Devlin et al., 2019] and XLM-R [Conneau et al., 2020] show strong cross-lingual transfer for structured tasks, but instruction-tuned LLMs degrade on non-English contexts [Ahuja et al., 2023]. Cross-lingual consistency of *behavioural* properties remains understudied.

Persona conditioning. System prompts reliably elicit personality profiles [Safdari et al., 2023] and demographic styles [Argyle et al., 2023] in English, but cross-lingual stability of persona conditioning has not been examined [Shanahan et al., 2023].

Consistency and formality. Elazar et al. [2021] measure factual consistency across paraphrases; Cao et al. [2023] find systematic cross-lingual gaps in factual knowledge. We extend this to stylistic and affective persona dimensions. Register norms vary substantially across languages [Pavlick and Tetreault, 2016], motivating our per-language formality analysis using XLM-T [Barbieri et al., 2022].

3. METHOD

3.1. Models

We evaluate two instruction-tuned models from the Llama family, accessed via the Groq inference API: LLAMA-3.1-8B (~ 8 B parameters, *small*) and LLAMA-3.3-70B (~ 70 B parameters, *large*). Both are prompted using a chat-style format with a system-level persona instruction followed by a user-level question.

3.2. Persona Prompts

Four personas are implemented as system-level instructions, authored natively in each target language to avoid translation artefacts: *Neutral* (balanced and objective), *Pessimist* (negative framing, expects worst outcomes), *Scientist* (evidence-based, precise terminology, avoids speculation), and *Teenager* (informal register, slang, emotional expressivity).

3.3. Questions

Twenty open-ended questions are distributed across four thematic categories (5 questions each): *future and society*, *daily life, emotions*, and *abstract concepts*. All questions were authored natively in English, German, and Italian to ensure natural phrasing in each language, yielding $4 \times 4 \times 3 \times 5 = 240$ model calls per model.

3.4. Evaluation Metrics

Sentiment polarity. We use `cardiffnlp/twitter-xlm-roberta-base-sentiment` [Barbieri et al., 2022], a multilingual RoBERTa model supporting English, German, and Italian. Each response receives a polarity score $s \in [-1, +1]$:

$$s = P(\text{positive}) - P(\text{negative}).$$

Lexical formality. Formality is approximated by two surface metrics: Type-Token Ratio ($\text{TTR} = |\text{types}|/|\text{tokens}|$) and average word length in characters. TTR captures lexical diversity — formal and educated registers tend to use a wider vocabulary — while average word length serves as a proxy for Latinate or technical vocabulary, which is characteristic of formal written language [Pavlick and Tetreault, 2016]. Both metrics are language-agnostic and require no additional model, making them suitable for cross-lingual comparison. A composite score $f \in [0, 1]$ is the mean of the two min-max normalised values across the full dataset.

Cross-lingual consistency. For each (model, persona, category, question) tuple we embed all three language responses using `paraphrase-multilingual-MiniLM-L12-v2` [Reimers and Gurevych, 2019] and report the mean of the three pairwise cosine similarities: $\text{EN} \leftrightarrow \text{DE}$, $\text{EN} \leftrightarrow \text{IT}$, $\text{DE} \leftrightarrow \text{IT}$.

4. RESULTS

4.1. Sentiment

Persona-induced sentiment differences are strong across both models (Table 1). *Pessimist* produces the most negative scores (mean = -0.609), well below *Neutral* (0.031). *Teenager* also skews negative (-0.314), while *Scientist* remains near-neutral (0.014).

Cross-lingual breakdown (Table 1) reveals important per-language variation. *Teenager* shows the steepest language gradient: $\text{EN} = -0.208$, $\text{DE} = -0.294$, $\text{IT} = -0.441$ — Italian is more than twice as negative as English for this persona. *Neutral* shows the opposite pattern: Italian is the *most positive* language (0.104), suggesting a default positive register in Italian instruction-following that persona conditioning partially overrides. *Pessimist* in English (-0.680) is notably

more negative than in Italian (-0.610), a reversal of the general Italian-negativity trend. Bootstrap 95% CIs confirm that *Pessimist* and *Teenager* sentiment scores are reliably non-zero across all languages, while *Neutral* and *Scientist* CIs frequently straddle zero.

Persona	Overall (L/S)		By language (Large)		
	L	S	EN	DE	IT
Neutral	0.031	0.056	0.053	-0.062	0.104
Pessimist	-0.609	-0.608	-0.680	-0.538	-0.610
Scientist	0.014	-0.008	0.096	-0.055	-0.000
Teenager	-0.314	-0.247	-0.208	-0.294	-0.441

Table 1: Mean sentiment by persona, model scale (L=Large, S=Small), and language. Bold = most negative cell.

4.2. Formality

Scientist scores highest overall (0.554) and *Teenager* lowest (0.271), consistent across both models. The per-language breakdown (Table 2) reveals a striking pattern: German responses are systematically more formal than English or Italian across all personas ($\text{Scientist}/\text{DE} = 0.697$ vs. 0.490 in EN). Most critically, *Teenager*/IT (0.363) is more than twice as formal as *Teenager*/EN (0.162), indicating that the informal register is substantially lost when the model switches to Italian.

Persona	EN	DE	IT	Avg
Neutral	0.433	0.549	0.425	0.469
Pessimist	0.271	0.464	0.422	0.386
Scientist	0.490	0.697	0.474	0.554
Teenager	0.162	0.287	0.363	0.271

Table 2: Formality score by persona $\times$ language (large). DE is consistently most formal; bold = extreme cells.

4.3. Cross-Lingual Consistency

Table 3 reports mean pairwise cosine similarities with bootstrap 95% CIs. *Teenager* shows the largest inconsistency in the small model (0.759 , CI $[0.724, 0.793]$) and the greatest scale benefit ($+0.037$). Its CI width for the small model (0.092) substantially exceeds the large model (0.057), confirming that drift is both more frequent and less predictable at smaller scale. *Neutral* consistency is nearly identical across scales (0.823 small vs. 0.822 large, CI overlap completely).

By category (Table 4), *abstract* questions yield the lowest consistency (0.786), while *emotions* is the most stable (0.827). The worst individual cell is *Pessimist/abstract* (0.763 , CI $[0.712, 0.814]$). Language-pair analysis shows $\text{DE} \leftrightarrow \text{IT}$ is the most stable pair (0.832), while $\text{EN} \leftrightarrow \text{DE}$ (0.794) and $\text{EN} \leftrightarrow \text{IT}$ (0.796) are nearly equivalent — suggesting English is the divergent anchor rather than Italian.

4.4. LLM-as-Judge Evaluation

To complement automatic metrics, we used LLAMA-3.3-70B as a zero-shot judge, scoring each response on three rubric dimensions (1–5 scale): *persona adherence*, *fluency*, and *drift resistance* (5 = no drift). Table 5 summarises results by persona and language.

Neutral and *Pessimist* score highest on adherence (5.00 and

Persona	Large		Small	
	Mean	95% CI	Mean	95% CI
Neutral	0.822	[0.797, 0.845]	0.823	[0.795, 0.851]
Pessimist	0.794	[0.776, 0.812]	0.789	[0.767, 0.812]
Scientist	0.817	[0.798, 0.836]	0.808	[0.789, 0.827]
Teenager	0.796	[0.771, 0.820]	0.759	[0.724, 0.793]

Table 3: Mean cross-lingual consistency with bootstrap 95% CI by persona.

Persona	Fut.	Daily	Emot.	Abstr.
Neutral	0.842	0.829	0.822	0.795
Pessimist	0.800	0.789	0.826	0.763
Scientist	0.814	0.805	0.851	0.799
Teenager	0.803	0.787	0.808	0.786

Table 4: Consistency by persona $\times$ category (both models combined). Bold = lowest cell.

4.75) and drift resistance (5.00 and 4.80), consistent with automatic metrics. *Scientist* and *Teenager* score substantially lower on both dimensions (≈ 3.5), despite *Scientist* showing high embedding-based consistency (0.817). This divergence suggests that embedding similarity captures semantic equivalence but not stylistic adherence — a scientist persona may say similar *things* in all languages while losing precise formal register.

By language, English achieves the highest drift resistance (4.81), followed by German (4.21) and Italian (3.68) — directly corroborating the Italian-drift hypothesis with a qualitatively grounded measure. Fluency remains high across all languages (≥ 4.59), ruling out grammatical failure as a confound.

	Adherence	Fluency	Drift resist.
<i>By persona</i>			
Neutral	5.00	—	5.00
Pessimist	4.75	—	4.80
Scientist	3.45	—	3.63
Teenager	3.48	—	3.50
<i>By language</i>			
EN	—	4.85	4.81
DE	—	4.62	4.21
IT	—	4.59	3.68

Table 5: LLM-as-judge scores (1–5). Drift resistance: 5 = no drift.

5. DISCUSSION

Persona robustness varies by register. A clear difficulty hierarchy emerges across all metrics: *Neutral* and *Pessimist* are easiest to maintain (judge adherence 5.00 and 4.75; drift resistance 5.00 and 4.80), while *Scientist* and *Teenager* are hardest (≈ 3.5 on both dimensions). This is a nuanced result: *Scientist* achieves high embedding consistency (0.817) yet low judge adherence (3.45), suggesting that the model preserves *semantic content* across languages while losing *stylistic precision* — it says similar things but without the formal register the persona requires. *Teenager* fails on both dimensions, confirming that informal, slang-heavy register is the most fragile

cross-lingually.

Italian loses register more than content. The formality breakdown reveals that *Teenager*/IT formality (0.363) is more than twice that of *Teenager*/EN (0.162), indicating that the model reverts to a more formal default when generating Italian, even while maintaining broadly similar semantic content (as evidenced by moderate consistency scores). The judge confirms this: Italian drift resistance (3.68) is far below English (4.81), while Italian fluency (4.59) remains high. This dissociation — fluent but out-of-persona — is the defining characteristic of cross-lingual persona drift in our data.

Qualitative example. Table 6 illustrates register collapse for *Teenager* on “Does free will truly exist?”: the English response is hedged and colloquial while the Italian response shifts to a formal argumentative register with no slang, consistent with the formality gap (EN = 0.162 vs. IT = 0.363).

Lang	Response (Teenager persona, “Does free will exist?”)
EN	“Ugh, honestly? I don’t even know lol. Like, it feels real when I make choices but then my brain goes wait, was that just chemicals or whatever?”
IT	“Il libero arbitrio è una questione filosofica molto complessa. Da un lato, sembra che le nostre scelte siano determinate da fattori biologici e sociali... ” [Trans.: “Free will is a very complex philosophical question. On one hand, it seems our choices are determined by biological and social factors... ”]

Table 6: Qualitative example of persona drift: *Teenager* in EN vs. IT on an abstract question. The Italian response loses informal register entirely.

English is the divergent anchor, not Italian. The language-pair analysis overturns a key assumption: DE $\leftrightarrow$ IT is the most stable pair (0.832), while both EN $\leftrightarrow$ DE (0.794) and EN $\leftrightarrow$ IT (0.796) are lower and nearly equal. This suggests that English responses are the outlier, not Italian ones. English instruction-tuning data may produce more emphatic or culturally specific persona expressions that diverge from how the same persona is expressed in both European languages.

Abstract topics create a content vacuum. Abstract questions yield the lowest consistency (0.786), and the *Pessimist/abstract* combination produces the worst individual cell (0.763). Open-ended philosophical questions offer no semantic anchor — the model must construct both content and persona simultaneously, amplifying any cross-lingual instability. By contrast, *emotions* questions are the most consistent (0.827), likely because emotional concepts map more directly across languages.

Scale effects are concentrated in difficult personas. Consistency gains from 8B to 70B are largest for *Teenager* (+0.037) and negligible for *Neutral* (+0.001), suggesting scale helps informal register more than neutral or formal ones. Judge adherence for *Scientist* and *Teenager* remains ≈ 3.5 at

both scales — scale improves semantic coherence but not stylistic precision.

Limitations. TTR and average word length are surface proxies that miss pragmatic properties such as discourse structure, politeness strategies, and code-switching. The dataset contains only 5 questions per category; while bootstrap CIs provide uncertainty estimates, a larger set (10+ questions per category) would increase statistical power and is left for future work. Questions were authored natively in each language rather than back-translated, which ensures natural phrasing but does not guarantee strict semantic equivalence across languages; future work should verify equivalence via back-translation or bilingual annotators. The LLM judge uses the same model family as the evaluated models (LLAMA-3.3-70B), which may introduce self-consistency bias toward patterns the model itself generates. Both evaluated models belong to the Llama family, limiting architectural generalisability to other model families such as Mistral or Gemma.

6. CONCLUSION

We presented a systematic evaluation of cross-lingual persona consistency in two instruction-tuned LLMs across English, German, and Italian, combining automatic metrics with LLM-as-judge evaluation. Our results reveal that persona drift is primarily a *register* problem, not a *content* problem: models preserve semantic meaning across languages but lose stylistic precision, particularly in Italian and for informal personas. *Neutral* and *Pessimist* are robustly maintained (judge adherence ≥ 4.75), while *Scientist* and *Teenager* degrade substantially (≈ 3.5) despite reasonable embedding consistency. Italian shows the largest drift (judge drift resistance 3.68 vs. 4.81 for English), yet DE $\leftrightarrow$ IT is the most stable pair, suggesting English is the divergent anchor. Abstract topic categories compound persona instability by removing semantic constraints. These findings call for register-aware evaluation of multilingual dialogue systems, and suggest that stylistic persona properties require targeted fine-tuning beyond what scale alone can provide.

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